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EXPERIMENT – 11

Performing IP Spoofing Using a GitHub Tool in Kali Linux (Educational Demonstration)

Aim

Todemonstrate IPspoofing techniquesusing an open-sourceGitHubtoolinKali Linux for understanding network-level attack concepts and security vulnerabilities.

Software Requirements

OperatingSystem: Windows / Linux / macOS
VirtualizationSoftware: Oracle VirtualBox
GuestOperatingSystem: Kali Linux
ToolSource: GitHub (IP Spoofing Tool – Educational Use)
ProgrammingLanguage: Python / Bash (as per tool)
SystemRequirements:

- RAM:Minimum 4 GB (8 GB recommended)
- Disk Space: Minimum 30 GB
- Processor: Intel/AMD with Virtualization support

Introduction

IP spoofing is a technique where an attacker forges the source IP address in network packets to impersonate another system. It is commonly used in attacks such as DDoS, session hijacking, and bypassing IP-based authentication. This experiment demonstrates

Procedure

Step 1: Start Kali Linux Virtual Machine

1. OpenOracleVirtualBox. 2. StartKaliLinux.

Step 2: Open Terminal

1. LaunchTerminalinKaliLinux.
2. Updatesystempackages:sudoaptupdate

Step 3: Clone IP Spoofing Tool from GitHub

1. Clonetherepository:
2. gitclonehttps://github.com/<username>/<ip-spoofing-tool>
3. Navigatetothetooldirectory:cdip-spoofing-tool

Step 4: Install Dependencies

1. Install required dependencies: sudo apt install python3
2. pip3 install -r requirements.txt

Step 5: Configure the Tool

1. Openthescriptfile:nanospoof.py
2. Configure:
   - o TargetIPaddress
   - o SpoofedsourceIPaddress

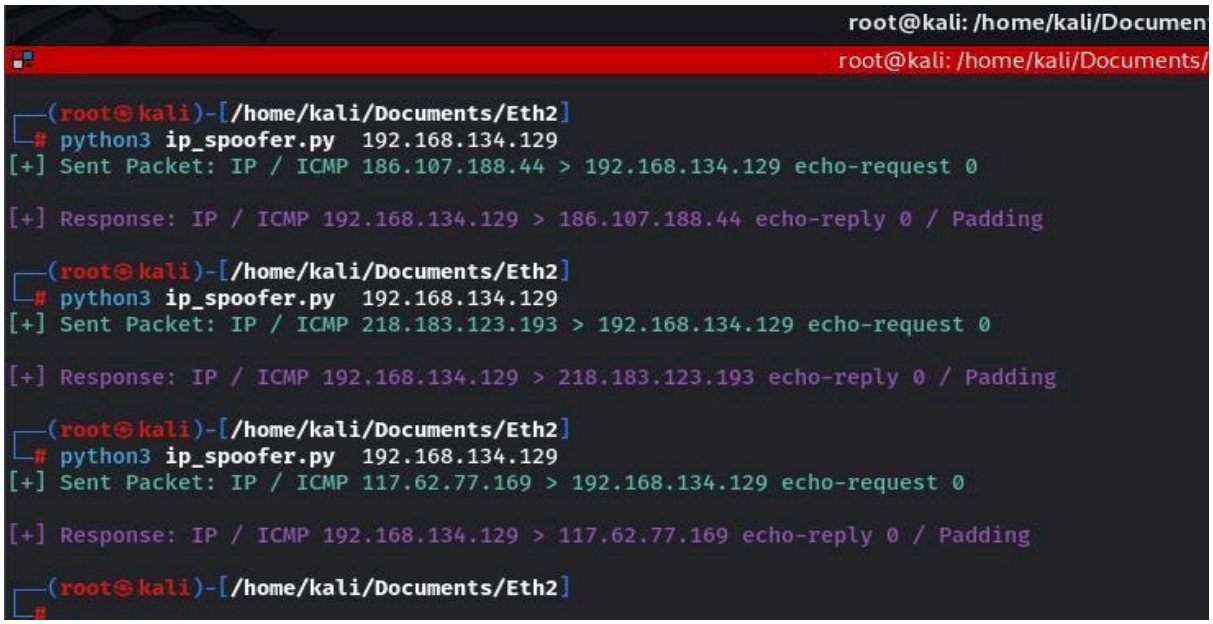
Step 6: Execute IP Spoofing

1. Runthetoolwithrootprivileges:sudopython3spoof.py
2. ObservepackettransmissionwithspoofedIPaddresses.

Step 7: Monitor Network Traffic (Optional)

1. UseWiresharkortcpdump:sudotcpdump
2. VerifyspoofedsourceIPpackets.

Output



Result

IP spoofing was successfully demonstrated using a GitHub-based tool in Kali Linux.
The experiment illustrates how attackers can manipulate IP headers and highlights the need for robust network security mechanisms.